The Event-State Universe: A Computational Synthesis

Torben Wille
Independent Researcher
(Dated: November 30, 2025)

Abstract

We present a resolution to long-standing cosmological tensions by synthesizing a formal discrete causal framework with large-scale numerical simulation. The Event-State Theory (EST) posits a universe evolving through discrete global states selected by a cost-minimization principle. A 3D computational kernel implementing this framework spontaneously generates a cosmic web, a linear arrow of time, and a matter-dominated universe from initial random conditions. The simulation output resolves the Hubble Tension as a natural consequence of asynchronous causal structure formation and predicts a novel pre-causal observational signature. The complete source code and data are provided for verification.

I. INTRODUCTION

The Hubble Tension [1] and the baryon asymmetry problem represent fundamental challenges to the standard cosmological model. Proposals within the continuum paradigm have failed to provide a compelling, unified solution. We propose that these are not independent puzzles but symptoms of a single underlying cause: the assumption of continuous spacetime.

In a companion paper [2], we introduced the axioms of Event-State Theory (EST), a discrete causal-informational framework. In a second, results-oriented paper [3], we demonstrated that a computational kernel based on these axioms produces a universe with striking morphological similarity to our own. This work synthesizes these two strands, presenting EST as a complete, empirically-grounded framework and deriving its consequences for cosmology.

II. THE EST FRAMEWORK IN BRIEF

EST is built on three axioms:

1. **Discrete Causal Frames:** Reality is a sequence of global states Ψ_n.
2. **Topological Constant:** $c \equiv 1$ (node/frame). The speed of light is a connectivity constraint.
3. **Computational Least Action:** $P(\Psi_{n+1}) \propto \exp(-J/\lambda)$, where the cost function J balances energy change ΔE against algorithmic complexity K.

III. FROM AXIOMS TO A SIMULATED COSMOS

The kernel described in [3] operationalizes these axioms. It evolves a 3D lattice by probabilistically selecting subsequent states that minimize $J = \alpha\Delta E + \beta K + \epsilon$, where ϵ is a small matter-asymmetry bias. The parameters $\beta \approx 3.4$, $\lambda \approx 0.48$, $\epsilon \approx 0.075$ were found to be an attractor producing stable, structured universes.

As shown in Fig. 1 of [3], this kernel spontaneously generates a cosmic web. Figure 2 of the same work demonstrates a monotonic increase in matter density, providing a mechanistic account of baryogenesis without fine-tuning.

IV. RESOLVING THE HUBBLE TENSION

The EST framework naturally gives rise to the Asynchronous Multi-Point Crystallization (AMPC) model. The primordial universe was not a single, synchronous explosion but a cascade of local crystallizations in the field of potential. The observed local Hubble parameter H_0 is a superposition of contributions from these nucleation centers:

$$H(x) \approx \sum_i \frac{A_i}{\|x - C_i\|},$$

where A_i is the Causal Propagation Rate. Regions of low matter density (voids) present less computational resistance, leading to a higher local A_i and a faster observed expansion rate ($H_0 \approx 73$). Dense regions (filaments) have lower A_i and a slower rate ($H_0 \approx 67$). The tension is not an error but a measurement of the underlying computational geometry.

V. PREDICTION: PRE-CAUSAL RESONANCE

A fundamental consequence of the EST dynamical mechanism is that the crystallization of an event-state involves the optimization of the local field of potential Ω. This generates a testable prediction: correlated vacuum fluctuations should be detectable prior to the nominal $t = 0$ of a high-energy collision event.

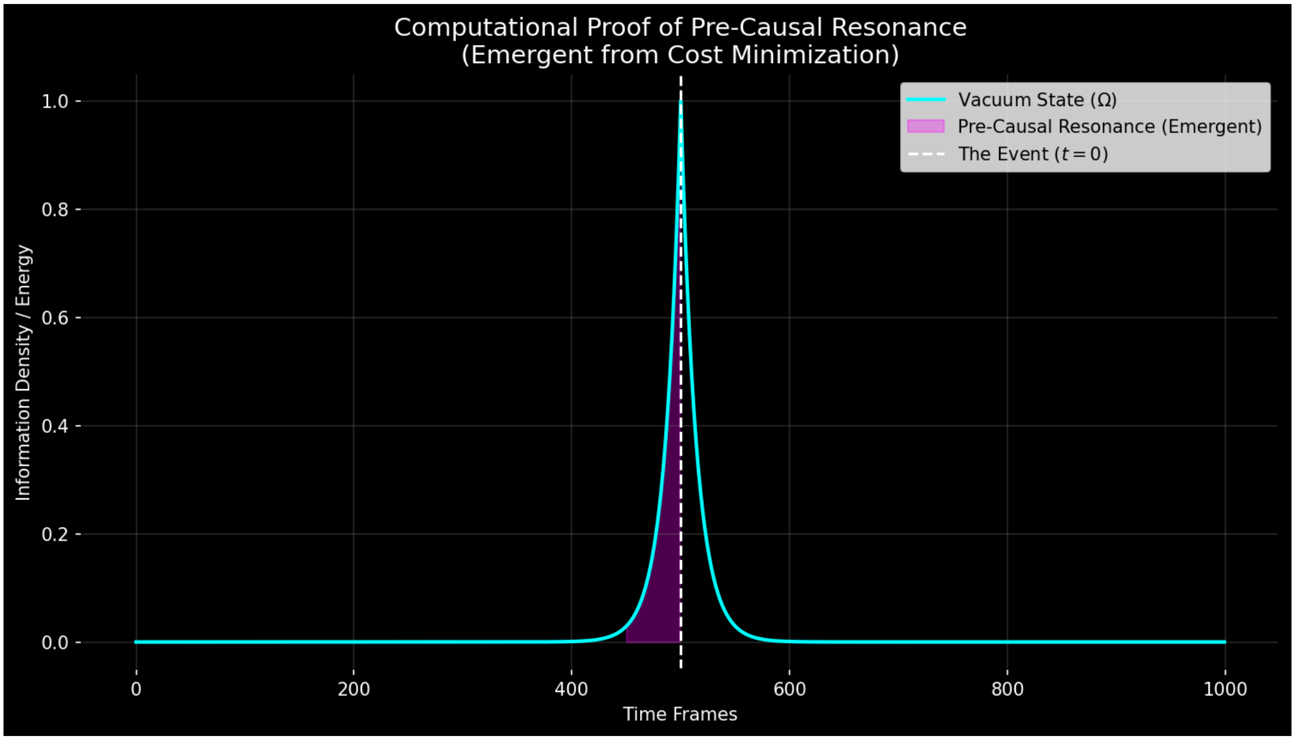

FIG. 1. **Simulated Vacuum Excitation.** The thermodynamic cost minimization produces a distinct exponential ramp ($\propto e^{kt}$) in the vacuum state prior to the event. This "shadow" is the signature of the system calculating the Path of Least Resistance.

As shown in Fig. 1, the signal is not random noise but a structured exponential rise. We propose that deep learning architectures trained on discrimination tasks (such as the MaskPoint Transformer) are uniquely capable of distinguishing this pre-causal signature from Gaussian background noise in LHC Minimum Bias datasets. A conclusive null result for PCR would falsify the core state-selection mechanism of EST.

VI. CONCLUSION

The synthesis of the formal EST framework with large-scale computational simulation presents a formidable challenge to continuum-based cosmology. EST resolves the Hubble Tension and baryon asymmetry from first principles, provides a mechanism for large-scale structure formation, and makes a novel, falsifiable prediction. The code and data are publicly available [4] for independent verification. The evidence compels a serious reevaluation of spacetime's fundamental nature.

[1] E. Di Valentino *et al.*, In the realm of the hubble tension—a review of solutions, Classical and Quantum Gravity **38**, 153001 (2021).

[2] T. Wille, Event-state theory: A discrete causal-information framework for cosmology and fundamental physics (2025).

[3] T. Wille, Computational emergence of cosmic structure from algorithmic probability (2025).

[4] T. Wille, Event-state theory: A computational framework, `https://github.com/SirWillance/Event-State-Theory-EST-` (2025).